

Signal Processing



Access and
acquire signals



Preprocess and
label signals



Transform signals
and extract features



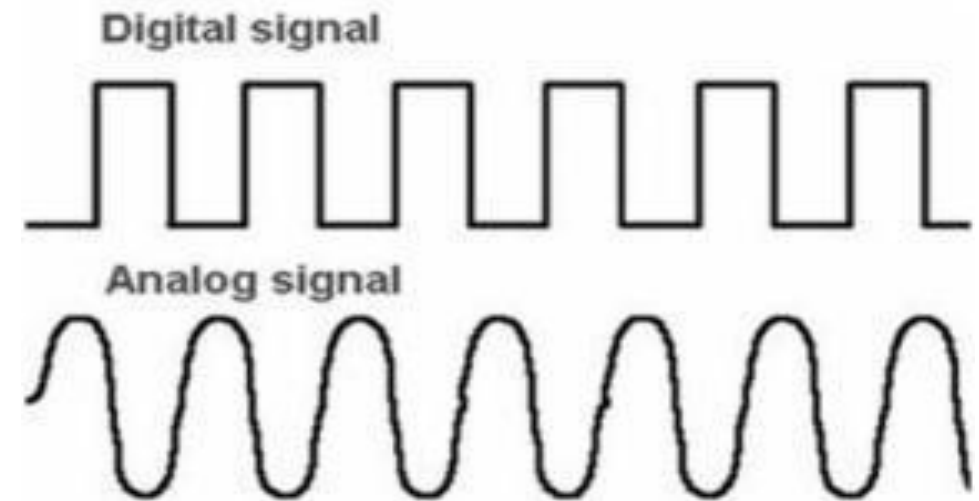
Train AI models



Deploy AI systems

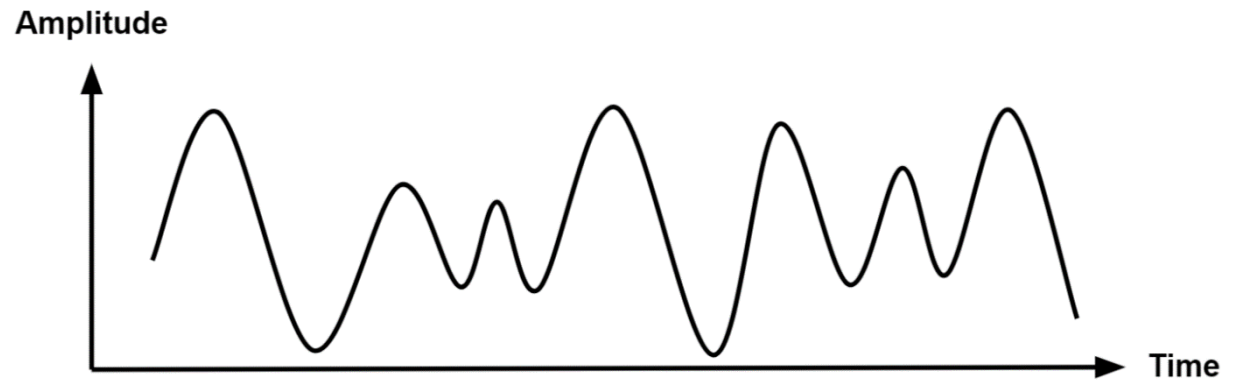
Analog Signal Vs Digital Signal

- **Analog** is the data signal in the form of a continuous wave, which carry information by changing the characteristics of a wave. Two parameters / the most important characteristic possessed by the analog signal is **amplitude and frequency**.
- **Digital** is a data signal in the form of pulses that can undergo sudden changes and has a **magnitude of 0 and 1**.



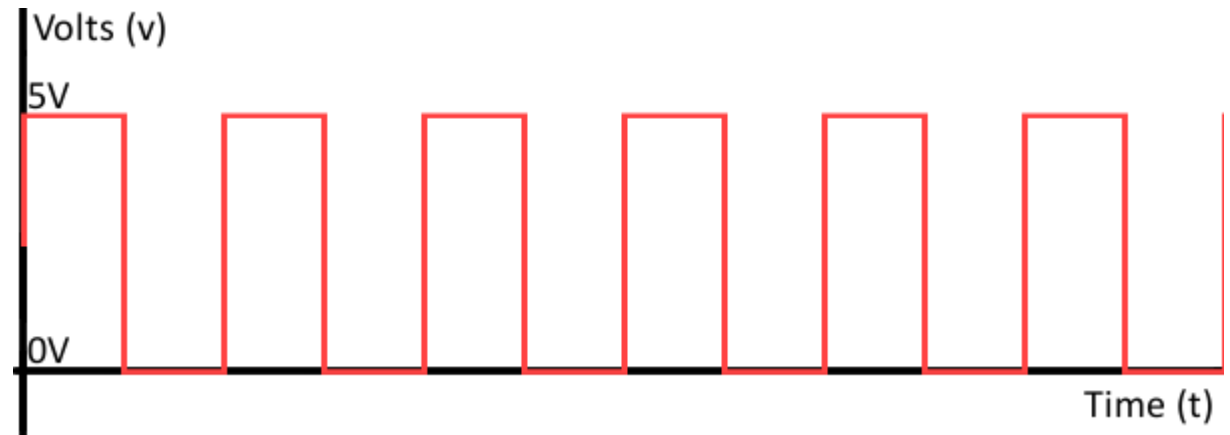
Analog Signal

- An **analog signal** is **continuous** in both time and amplitude.,
- **Examples:**
 - Human voice
 - Temperature readings
 - ECG signals
 - Radio waves
- **Characteristics:**
 - Infinite possible values
 - Smooth waveform
 - More sensitive to noise
 - Harder to store and process directly in computers



Digital Signal

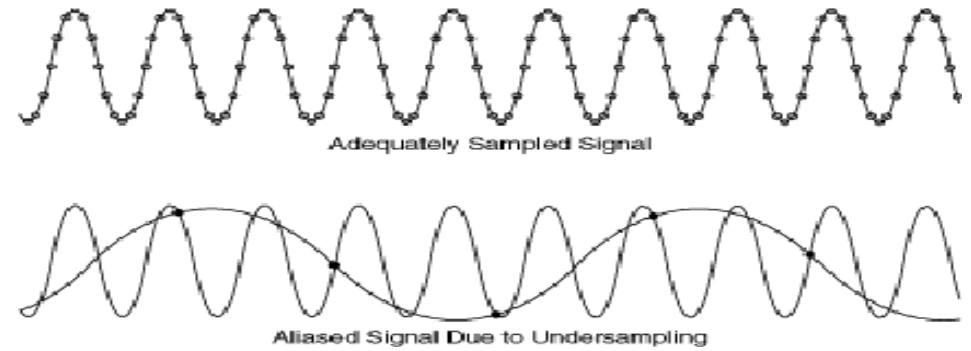
- A **digital signal** is **discrete** in time and amplitude, Takes specific values (usually 0 and 1), Step-like waveform, Used in computers
- **Examples:**
 - Binary data (0 and 1)
 - Digital images
 - MP3 files
 - Computer memory data
- **Characteristics:**
 - Finite values
 - Less sensitive to noise
 - Easier storage and transmission
 - Requires sampling & quantization



Preprocessing

Step	Purpose
Sampling	Convert to digital
Filtering	Remove noise
De-trending	Remove drift
Normalization	Scale data
Segmentation	Prepare for ML
Feature Extraction	Reduce dimensionality
Artifact Removal	Clean biomedical noise

Sampling



- Converting continuous-time signal into discrete-time signal.
- Sampling frequency f_s According to the **Harry Nyquist** theorem:

$$f_s \geq 2f_{max}$$

- **Aliasing happens when we sample a signal too slowly.**
If the sampling rate is not high enough, the signal looks like a **different (wrong) lower-frequency signal** after digitization.
- **How to Prevent Aliasing**
 - Use high enough sampling rate
 - Apply an **anti-aliasing low-pass filter** before sampling

Filtering

- Filtering is the process of selectively enhancing or removing certain components of a signal or image based on specific criteria, usually related to frequency or value.
- **Purpose:**
 - To remove unwanted components (e.g., noise, irrelevant features).
 - To extract useful features for analysis.

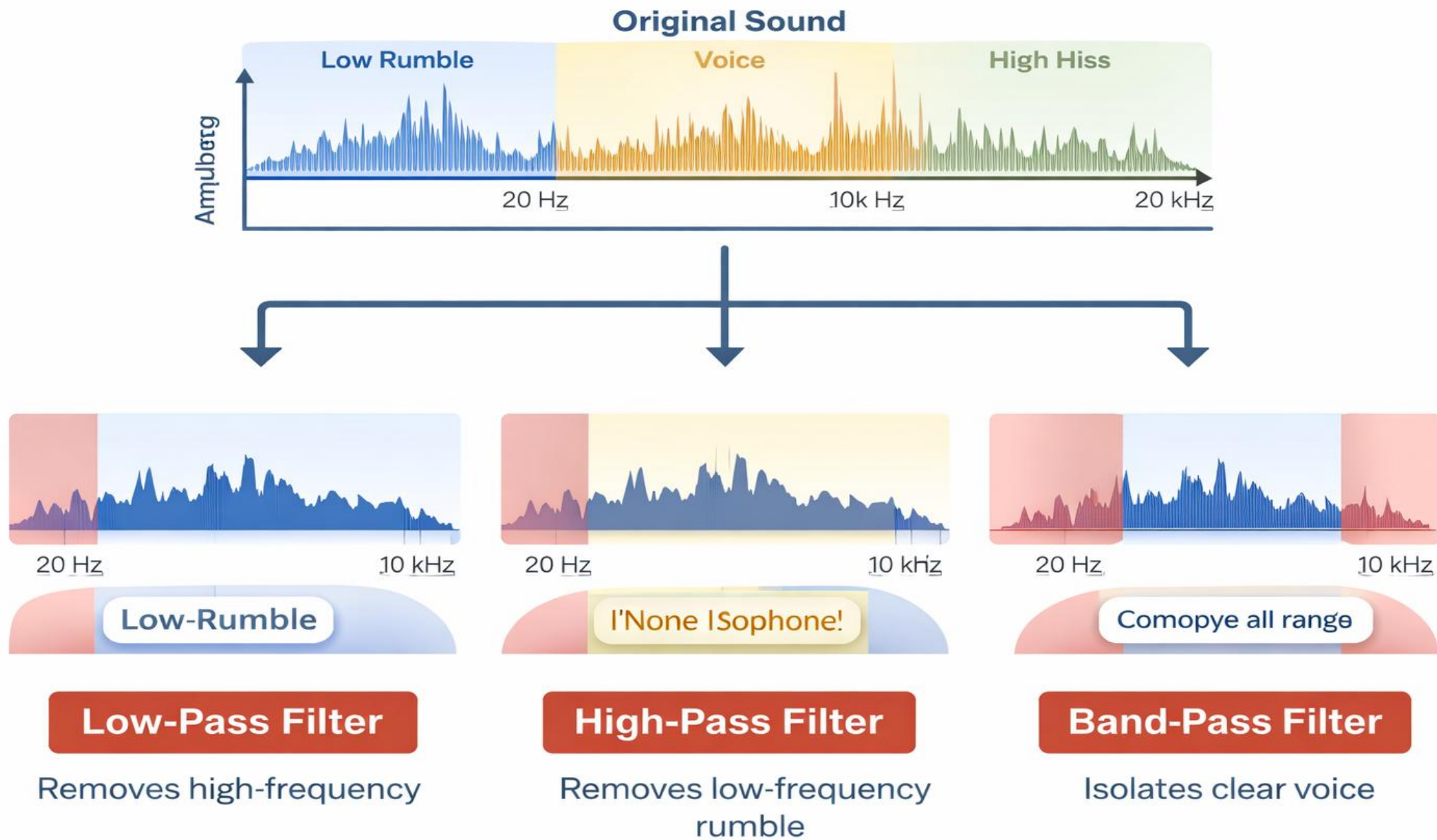
Filtering

- **Types of Filtering:**

- **Low-pass filter:** Allows low-frequency components to pass, blocks high-frequency components. Example: Smoothing an image or removing sudden spikes in sensor data.
- **High-pass filter:** Allows high-frequency components to pass, blocks low-frequency. Example: Edge detection in images.
- **Band-pass filter:** Allows only a specific range of frequencies. Example: Audio equalizers.

- **Example:**

- In image processing, a **Gaussian filter** smooths an image by reducing high-frequency noise.
- In audio, a **low-pass filter** can remove the hiss from a recording.



Normalization

Normalization is the process of scaling data or signals so that they lie within a certain range or have certain statistical properties.

Purpose:

- Standardize data for better comparison.
- Improve convergence in machine learning models.
- Reduce bias due to varying scales.

Normalization

Types:

- Min-max normalization: Scales data to a range, typically $[0, 1]$.
- Z-score normalization (standardization): Converts data to have a mean of 0 and standard deviation of 1.

Example:

- Image pixel values normalized from $[0, 255]$ to $[0, 1]$ before feeding into a CNN.
- Sensor data with different units is scaled to the same range before input into a model.

De-trending

- **Definition:**

De-trending is the process of **removing a trend (long-term change) from a signal** so that we can better analyze the underlying fluctuations or patterns. A “trend” is usually a slow increase or decrease over time that is not part of the high-frequency variations we want to study.

Segmentation

- **Definition:**

Segmentation is the process of **dividing a signal into meaningful parts or segments** based on certain criteria. Each segment represents a distinct feature, pattern, or state within the signal.

Raw Signal

|

v

Pre-processing (Filtering, Noise Reduction, Normalization)

|

v

Segmentation (optional)

|

v

Feature Extraction

| --- Time-domain features (mean, peak, etc.)

| --- Frequency-domain features (Fourier Transform → dominant frequency,
spectral energy)

|

v

Analysis / Machine Learning

Fourier Transformation

The **Fourier Transform (FT)** is a mathematical technique used to convert a signal from the **time domain** (how a signal changes over time) into the **frequency domain** (which frequencies make up the signal). It helps analyze complex signals by breaking them down into **simpler sine and cosine waves**.

Basic Idea: Decomposing a Signal

Any signal can be thought of as a combination of simple **sine and cosine waves** of different frequencies and amplitudes. The Fourier Transform helps determine:

1. **Which frequencies** are present in the signal.
2. **How strong each frequency** is (amplitude)?
3. **Phase shift** (how the wave is aligned in time).

Mathematically, the **Fourier Transform** of a function $f(t)$ is given by:

Where:

- $f(t)$ is the original function (time-domain signal).
- $F(\omega)$ is the transformed function (frequency-domain representation).
- $e^{-j\omega t}$ represents a complex exponential function (a combination of sine and cosine).
- ω is the angular frequency.

$$F(\omega) = \int_{-\infty}^{\infty} f(t)e^{-j\omega t} dt$$

Types of Fourier Transforms

- **Continuous Fourier Transform (CFT):**

- Used for continuous signals (e.g., sound waves).
- Converts an infinite-time signal into an infinite-frequency spectrum.

- **Discrete Fourier Transform (DFT):**

- Used for digital signals (e.g., audio stored in a computer).
- Works with a **finite** number of samples instead of an infinite function.
- The **Fast Fourier Transform (FFT)** is an efficient algorithm to compute the DFT quickly.

- **Inverse Fourier Transform (IFT):**

- Converts frequency-domain data **back** into the time domain.
- The Fourier Transform converts it into a set of **sinusoids (sine waves)**, each with a different **frequency, amplitude, and phase**. But what if you want to listen to the original song again? You need to **combine all those sinusoids back**—this is what the **Inverse Fourier Transform** does!
 - **Fourier Transform (FT)** → Breaks down a signal into frequencies.
 - **Inverse Fourier Transform (IFT)** → Reconstructs the signal from those frequencies.

Fourier Transform (FT)

- The Fourier Transform converts a **time-domain signal** (raw audio waveform) into the **frequency domain**.
- Essentially, it tells you **which frequencies are present** and **how strong they are**, but **without time localization** (or with limited time info if you do Short-Time FT).
- **Key Points:**
 - Input: Raw audio signal.
 - Output: Frequency spectrum (amplitude vs frequency).
 - Limitation: Standard FT gives global frequency content — you **lose information about when** a frequency occurs in time.
 - Solution for time info: **Short-Time Fourier Transform (STFT)** — breaks audio into small overlapping windows to see frequency changes over time.

Mel Spectrogram

- The Mel Spectrogram is **based on the STFT**, but it **maps frequencies onto the Mel scale**, which approximates **human hearing perception**.
- Humans are better at detecting small differences in low frequencies than in high frequencies. The Mel scale compresses high frequencies and expands low frequencies accordingly.
- **Key Points:**
 - Input: Usually the STFT of audio (or raw audio if using a library that computes both internally).
 - Output: 2D matrix of **time vs Mel-scaled frequency**, with intensity showing **amplitude**.
 - Advantage: Better representation for **machine learning on audio**, especially speech and music.
 - Optional: Can be further transformed into **MFCCs** for dimensionality reduction.

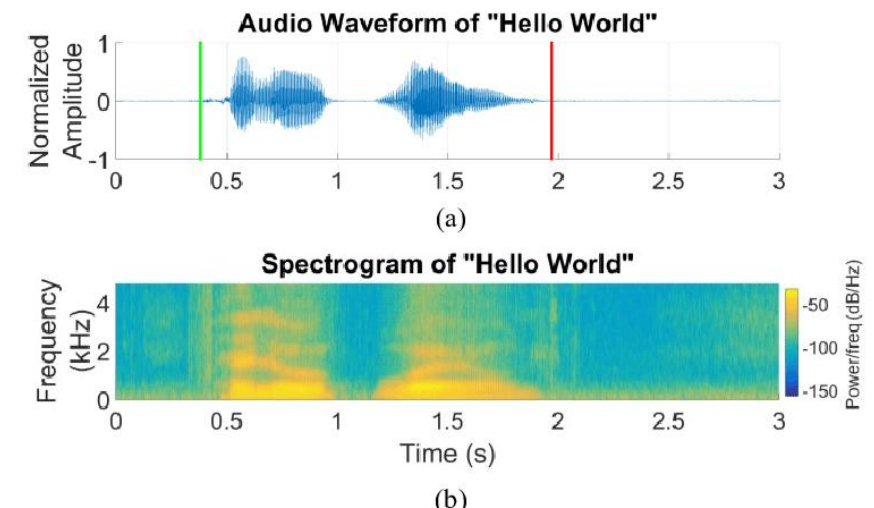
Mel Spectrogram:

- A **Mel spectrogram** is a way to represent sound in a way that matches how humans hear it. It is widely used in **speech recognition, music analysis, and machine learning applications**.

What is a Spectrogram?

- A **spectrogram** is a visual representation of sound. It shows how the frequencies in a signal change over time.
 - **X-axis (horizontal):** Time (how the sound evolves).
 - **Y-axis (vertical):** Frequency (pitch of the sound).
 - **Color/Intensity:** Amplitude (loudness of each frequency).

Example: If you say "hello," a spectrogram will show which frequencies were active at each moment of your speech.



What is the Mel Scale?

- The **Mel scale** is a way of mapping frequencies to match how humans perceive sound.
- Humans **hear lower frequencies (e.g., 100 Hz vs. 200 Hz) more distinctly** than higher ones (e.g., 5000 Hz vs. 5100 Hz).
- The **Mel scale** spaces frequencies so that equal changes in Mel correspond to **equal perceived changes in pitch**.

◆ Formula:

$$Mel(f) = 2595 \log_{10} \left(1 + \frac{f}{700} \right)$$

This warps high frequencies so that they get less emphasis, mimicking human perception.

How is a Spectrogram Created?

- **Convert the audio signal** from time domain to frequency domain using the **Fourier Transform (FT)**.
- **Apply the Short-Time Fourier Transform (STFT)** to analyze how frequencies change over time.

$$STFT(x(t)) = \sum_{n=0}^{N-1} x(n)w(n)e^{-j2\pi fn}$$

- • The **STFT** applies the **Fourier Transform** to **small segments (windows)** of the signal.
- • This captures **both time and frequency information**.

How is a Mel Spectrogram Created?

- **Step 1: Compute the Spectrogram (STFT)**
 - The **Short-Time Fourier Transform (STFT)** is applied to analyze frequencies over time.
- **Step 2: Apply Mel Filter Banks**
 - Instead of using **linear frequency bins**, we use **Mel-scaled frequency bins**.
 - A **Mel filter bank** is a set of overlapping triangular filters that convert frequencies to the Mel scale.
 - **Each filter focuses on a different frequency range.**
 - **Lower frequencies have finer resolution**, while higher frequencies are more compressed.

How is a Mel Spectrogram Created?

- Mathematically, a **Mel filter bank** consists of **triangular filters** that sum power in different frequency bands:

$$H_m(f) = \begin{cases} 0, & f < f_{m-1} \\ \frac{f - f_{m-1}}{f_m - f_{m-1}}, & f_{m-1} \leq f \leq f_m \\ \frac{f_{m+1} - f}{f_{m+1} - f_m}, & f_m \leq f \leq f_{m+1} \\ 0, & f > f_{m+1} \end{cases}$$

where f_m are the Mel-scaled center frequencies of the filters.

How is a Mel Spectrogram Created?

- **Step 3: Convert to Log Scale (dB Scale)**

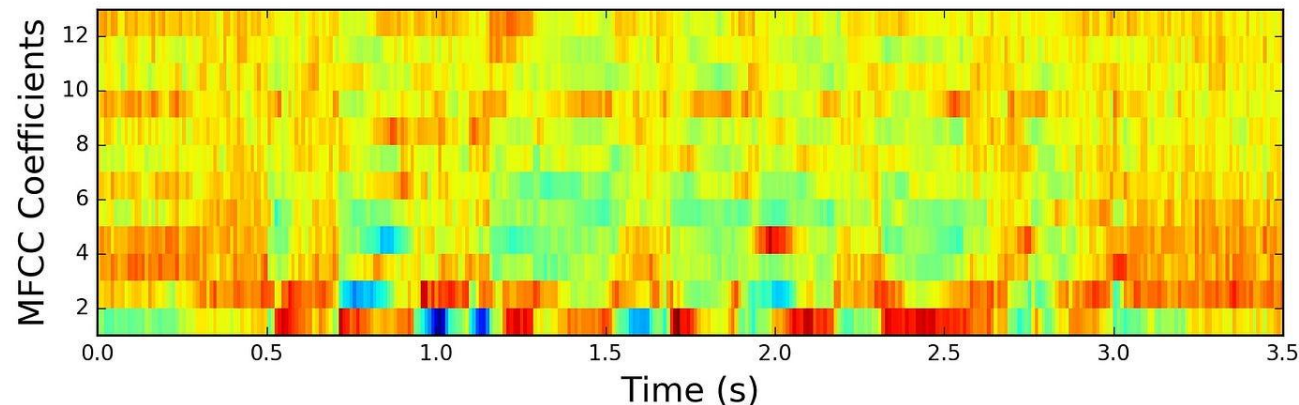
- After filtering, the Mel spectrogram is converted to a **logarithmic scale (decibels, dB)**:
- *“**Decibel (dB)** is a unit used to measure the **intensity of sound** or the **ratio of two values** (such as power or amplitude). It is a **logarithmic** unit, meaning it represents values on an exponential scale rather than a linear scale.”*

$$S_{mel} = \log(S)$$

- This **mimics how human ears perceive loudness**, as we perceive **loudness logarithmically** rather than linearly.

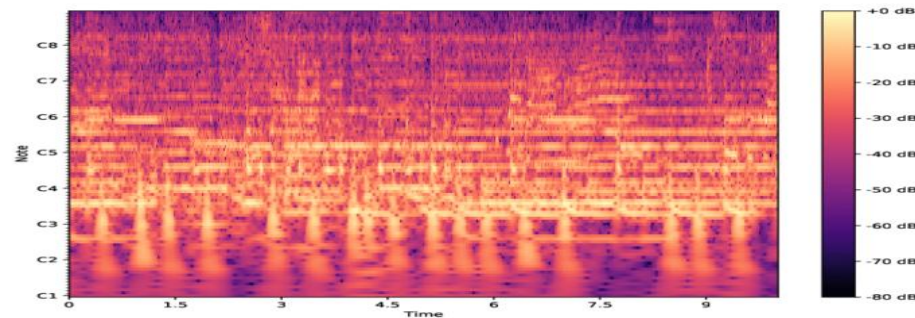
1. MFCC (Mel-Frequency Cepstral Coefficients)

- Derived from Mel Spectrogram
 - Compressed frequency representation using DCT
 - X-axis: Time
 - Y-axis: Cepstral Coefficient Index
 - Commonly used in speech recognition
 - Lightweight and computationally efficient



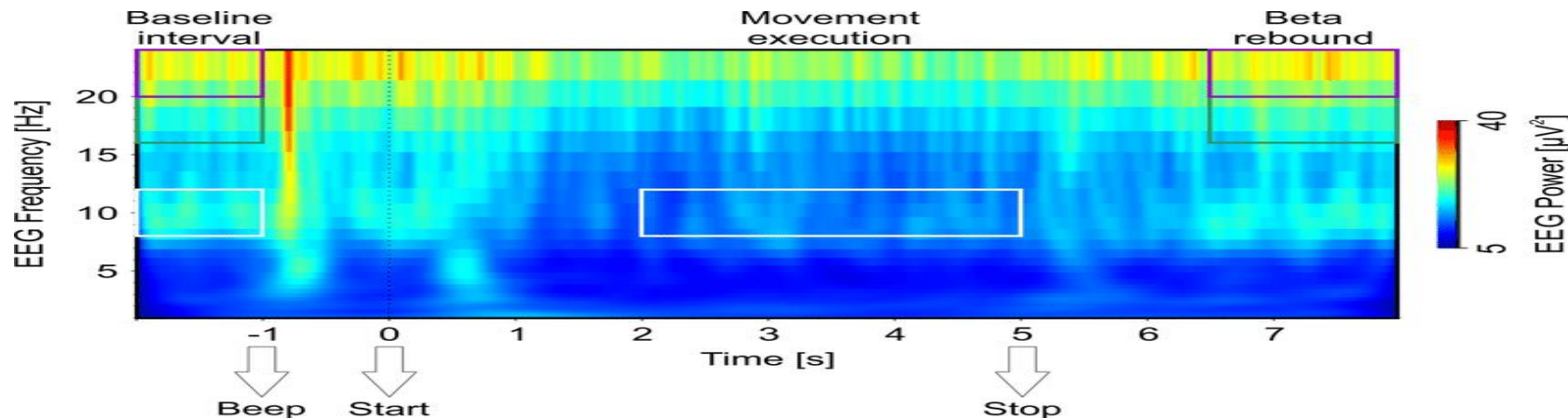
2. Constant-Q Transform (CQT)

- Logarithmically spaced frequency bins
 - Better resolution at low frequencies
 - Suitable for music analysis
 - X-axis: Time
 - Y-axis: Log-Frequency
 - Good for instrument and genre classification



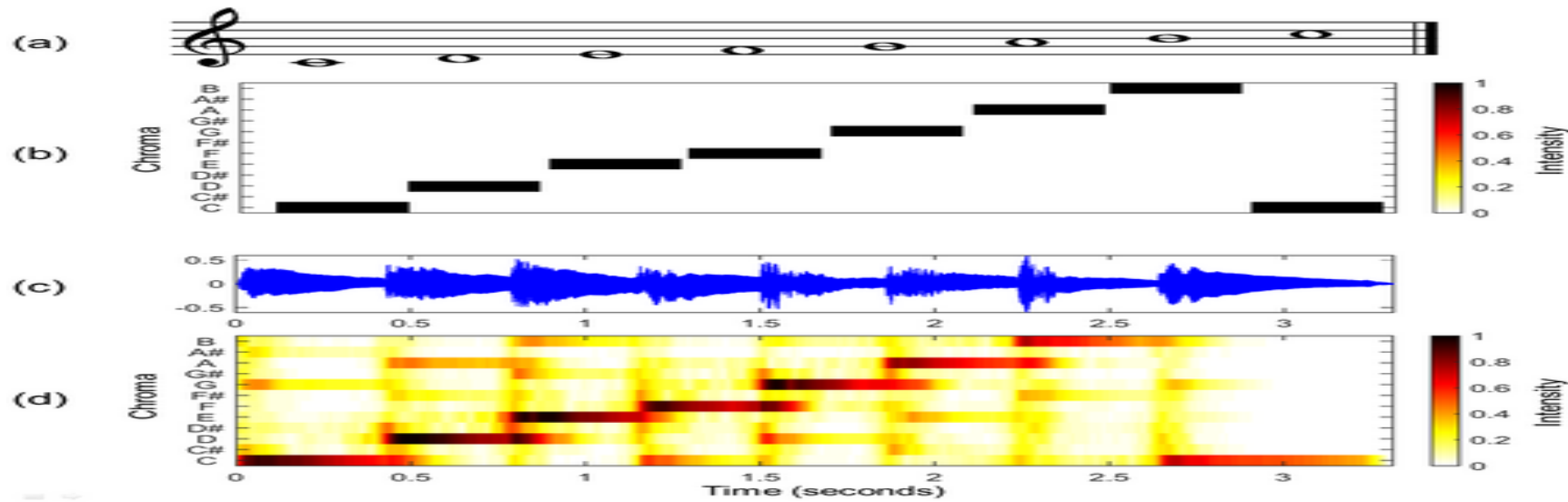
3. Wavelet Scalogram

- Based on Continuous Wavelet Transform
 - Better time localization than STFT
 - Suitable for non-stationary signals
 - X-axis: Time
 - Y-axis: Scale (related to frequency)
 - Common in EEG and biomedical analysis



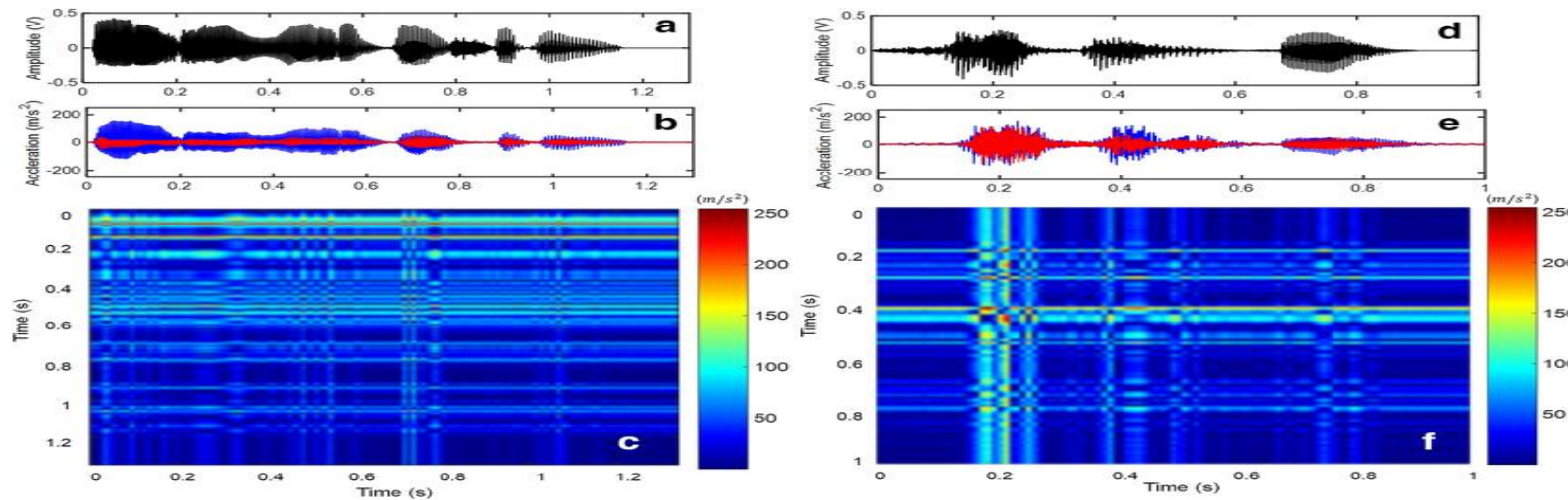
4. Chroma Features

- Represents 12 pitch classes (C, C#, D, etc.)
 - Compact harmonic representation
 - X-axis: Time
 - Y-axis: Pitch Class
 - Used in music information retrieval
 - Focuses on harmony rather than full spectrum



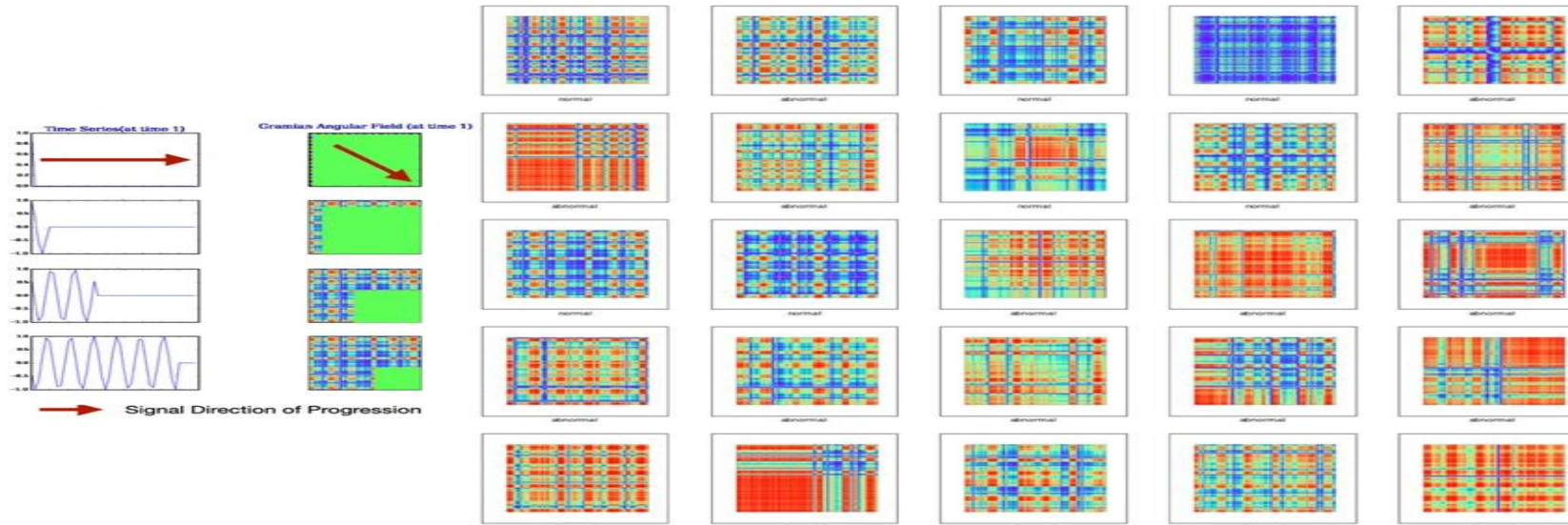
5. Recurrence Plot

- Square similarity matrix
 - Both axes represent time
 - Shows similarity between time segments
 - Useful for nonlinear systems
 - Applied in biomedical and dynamic systems analysis
 - Produces strong image-like patterns



6. Gramian Angular Field (GAF)

- Encodes time series using polar coordinates
 - Transforms 1D signal into 2D matrix
 - Captures temporal correlations
 - Produces structured texture patterns
 - Popular in time-series CNN research



When It Is Used

Technique	When It Is Used	Typical Applications
Gramian Angular Field (GAF)	Used when you want to convert a time-series signal into an image while preserving temporal correlations.	Time-series classification, anomaly detection, ECG analysis, sensor data analysis
Recurrence Plot (RP)	Used to visualize repeating patterns or periodic behavior in a signal by showing when a system revisits similar states.	Nonlinear time-series analysis, speech signals, physiological signals (EEG, ECG), dynamical system analysis
Chroma Features	Used when analyzing musical content , especially the pitch classes (12 semitones) in music regardless of octave.	Music genre classification, chord recognition, music similarity detection

When It Is Used

Technique	When It Is Used	Typical Applications
Wavelet Scalogram	Used when signals contain time-varying frequencies and you need good time–frequency localization . Works well for non-stationary signals .	Speech processing, biomedical signals, vibration analysis, fault detection
Constant-Q Transform (CQT)	Used when analyzing music or harmonic signals where frequency resolution should match musical pitch perception (logarithmic scale).	Music transcription, pitch detection, music classification
MFCC (Mel-Frequency Cepstral Coefficients)	Used to extract compact audio features that represent the human auditory perception of sound . One of the most common speech features.	Speech recognition, speaker identification, emotion detection, audio classification

